

- 1 Miss Ma wants to prepare a special diet for her pet called Rusty. She formulates a food diet consisting of food A, B and C that contains 110 grams of protein, 85 grams of carbohydrates and 45 grams of fat. The food's composition is shown in the following table.

Food	Percentage composition (%)			Unit Price \$/gram
	Protein	Carbohydrate	Fat	
A	60	30	10	0.60
B	40	30	30	0.40
C	30	70	0	0.30

How many grams of each food should Miss Ma use to prepare the food diet? [3]

What is the amount that Miss Ma has to spend on the food diet for Rusty? [1]

- 2 The functions f and g are defined by

$$f : x \rightarrow x^2 - 1, \quad x \in \mathbb{R}$$

$$g : x \rightarrow \sqrt{x+4}, \quad x \in \mathbb{R}^+$$

(i) Show that the composite function fg exists and define fg in similar form. [2]

(ii) If h is a function defined such that $h(x) = gfg(x)$ for all $x > 0$, show that the composite function gfg exists. Hence, find an expression for $h(x)$. [2]

(iii) Solve $h(x) = g^{-1}g(x)$, giving your answer in exact form. [2]

- 3 Lines l_1, l_2 are given by

$$\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \quad \text{and} \quad \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 7 \\ 5 \\ 1 \end{pmatrix} \quad \text{respectively.}$$

(i) Show that l_1 and l_2 are skew lines. [3]

(ii) Find the shortest distance from the point $A(1, 2, -1)$ to line l_1 . [3]

- 4 Given that $y = e^{\sin^{-1} x}$, show that $(1-x^2)\left(\frac{dy}{dx}\right)^2 = y^2$. [1]

By further differentiation of this result, find the Maclaurin's series for y in ascending powers of x up to and including the term in x^3 . [4]

Deduce the Maclaurin's series expansion of $y = \sin^{-1} x$ up to term in x^3 . [3]

- 5 (i) Find the root α , of the equation $f(x) = 0$, where $f(x) = x^3 + 3x - 10$. [1]

A sequence of real numbers $x_1, x_2, x_3, \dots$ satisfies the recurrence relation

$$x_{n+1} = (10 - 3x_n)^{\frac{1}{3}}$$

for $n \geq 1$. [2]

(ii) Prove algebraically that, if the sequence converges, then it converges to α .

(iii) Show that $x_{n+1} > x_n$ if $0 < x_n < \alpha$; [3]

$$x_{n+1} < x_n \text{ if } x_n > \alpha.$$

(iv) Another sequence v_n which satisfies the relation $v_{n+1} = \frac{2v_n^3 + 10}{3v_n^2 + 3}$ is used

to estimate α . By considering the equation of tangent at $(v_n, f(v_n))$, obtain the above recurrence relation, if the tangent cuts the x -axis at v_{n+1} . [3]

- 6 By considering $\frac{r-1}{r(r+1)} - \frac{r}{(r+1)(r+2)}$, express the series

$$\frac{2}{3 \cdot 4 \cdot 5} + \frac{4}{4 \cdot 5 \cdot 6} + \frac{6}{5 \cdot 6 \cdot 7} + \dots + \frac{2(N-2)}{N(N+1)(N+2)}$$

in the form $\frac{1}{3} - f(N)$. [4]

(i) Deduce that $\sum_{r=1}^{N-2} \frac{r}{(r+2)(r+3)(r+4)} < \frac{1}{6}$ for all $N \geq 3$. [3]

(ii) Given that $u_n = u_{n-1} - \frac{2(n-2)}{n(n+1)(n+2)}$ where n is a positive integer and

$u_1 = \frac{1}{6}$, find an expression for u_n in terms of n . [3]

7 The curve C has equation $y = \frac{2x^2 + 4}{3(x + \lambda + 1)}$.

It is given that C has a vertical asymptote $x = 0$.

(i) Determine the value of λ . [1]

(ii) Find the equation of the other asymptote of C . [2]

(iii) Prove, using an algebraic method, that C cannot lie between $-\frac{4\sqrt{a}}{3} < y < \frac{4\sqrt{a}}{3}$, $a > 0$, where a is to be determined. [3]

(iv) When $\lambda = 0$, the graph of $y = \frac{2x^2 + 4}{3(x + \lambda + 1)} + k$ cuts the x -axis at 2 distinct points where k is a positive integer.

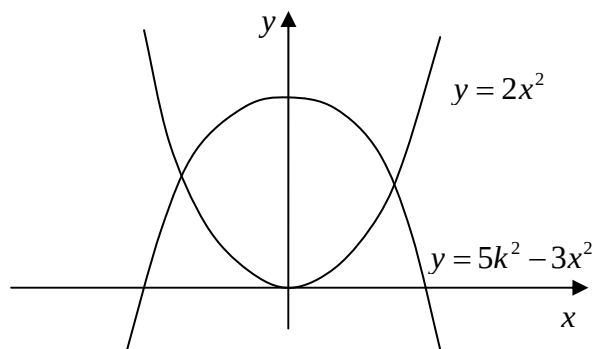
(a) Sketch the graph of $y = \frac{2x^2 + 4}{3(x + \lambda + 1)} + k$ when $\lambda = 0$, indicating clearly all asymptotes and coordinates of the axial intercepts. [4]

(b) Deduce the least possible integer of k . [1]

8 (a) Find $\int x \tan^{-1}(2x^2) dx$. [3]

(b) Given that $3 \int_0^m \frac{1}{\pi \sqrt{1 - 9x^2}} dx = \frac{1}{4}$, where $m > 0$. Find the exact value of m . [3]

(c) The diagram below shows the graphs of $y = 2x^2$ and $y = 5k^2 - 3x^2$, where k is a positive constant. Find, in terms of k , the volume of the solid of revolution obtained, when the region between the two curves is rotated through 2π radians about the x -axis.



[5]

- 9 (a)** In a convergent geometric progression, the sum of the first n terms is equal to the sum of the remaining terms. Given also that the $(n+1)$ th term is $\frac{1}{2}$, determine the value of the $(2n+1)$ th term. **[4]**

- (b)** An arithmetic progression has first term a and common difference d . The eighth, third and second term of the progression are successive terms of an infinite geometric progression.

If the first term of the geometric progression is 10, find the sum of the even-numbered terms of the progression. **[4]**

A sequence is formed in which the n th term is given by $\ln|u_n u_{n+1}|$, where u_n is the n th term of the geometric progression. Show that the sequence forms an arithmetic progression and state the value of the common difference. **[3]**

- 10 (i)** Write down the roots of the equation $w^3 = -i$. **[2]**

- (ii)** Hence, show that the values of z satisfying the equation $(i-z)^3 = iz^3$ are of the form

$$z = \frac{i}{1 - e^{\frac{1}{6}k\pi i}}, \quad \text{and state the values of } k \text{ that give all the roots of the equation.} \quad \text{[4]}$$

- (iii)** Express each root of $(i-z)^3 = iz^3$ in exact trigonometric form. **[6]**

- 11 (a)** A curve has parametric equations given by $x = 2t^2 + t$, $y = t^3 - 3t$.
- (i)** Sketch the curve, showing the exact coordinates of the axial intercepts. [2]
- (ii)** Find the equations of the tangents on the curve which are parallel to the y -axis. [3]
- (iii)** Find the coordinates of the point when the tangent at $t = -\frac{1}{2}$ meets the curve again. [3]
- (b)** A conical water tank with vertex down has a radius of 10 m at the top and is 24 m high. If water flows into the tank at a rate $20 \text{ m}^3\text{s}^{-1}$, how fast is the depth of the water increasing when the water is 16 m high? [4]

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